

24-HOUR LLM BUILD SPRINT

Hack Armenia

Build an LLM system that can **evaluate** evidence, **reason** under uncertainty, and **generalize** — for Armenia's public interest.

August 8–9, 2026 · AI9 Startup Campus · Yerevan

Welcome to Hack Armenia

One weekend. One theme. A working LLM system that actually does something useful for Armenia — and evidence that it works.

🏗️

Build

24 hours to go from idea to a running prototype, with mentors in the room.

📊

Measure

Show numbers. An evaluation — even a lightweight one — is what separates the top teams.

🎤

Pitch

5 minutes in front of an expert jury on Sunday. Talk plus live demo.

The sprint in numbers

24h
of building
Sat 10:00 → Sun

60–80
participants
one cohort

15–20
teams
3–4 people each

12
mentors
on-site Saturday

9
jury members
judging Demo Day

9
domains
Armenia's public interest

The theme: Armenia's public interest

Nothing is revealed at kickoff — the theme *is* the brief. Pick a real problem that matters here, and build for it.

Evaluate

Weigh evidence and cite what it is grounded in.

Reason

Work through uncertainty instead of guessing confidently.

Generalize

Hold up across real cases, not just your happy path.

The domains and example projects that follow are there to guide you — bring your own idea in the same spirit.

Nine public-interest domains

Where LLMs can be genuinely useful to Armenia. Choose one — or bring your own.



Urban development
City ops, planning & mobility



Public systems & policy
Filings, evidence & transparency



Healthcare
Safe triage, admin & patient education



Education
Tutoring, grading & Armenian-language learning



Security
Phishing defense, secure code & threat triage



Food & agriculture
Crop advisory, food safety & nutrition



Environment & energy
Efficiency, hazards & resources



Finance & fintech
Fair credit, fraud & literacy



Emergency & disaster
Coordination, preparedness & response

Example projects

Starting points, not requirements. Full list on the website.

🏛️

SME tax-filing copilot

Walk a small business through declarations; extract figures from receipts and pre-fill the forms.

🩺

Guarded symptom triage

Route symptoms to the right level of care in Armenian — with hard safety rails. Not diagnosis.

🎓

Auto-grading assistant

Grade short answers against a rubric, write feedback, and flag borderline cases for the teacher.

🔒

Phishing & scam detector

Catch bank-impersonation SMS and fake government notices aimed at Armenian users.

🍎

Crop advisory agent (VLM)

Diagnose plant disease from a photo and advise on treatment and timing, in Armenian.

🚨

Response coordination agent

Turn chaotic incoming incident reports into a de-duplicated, prioritised response picture.

Rules & tools

🧠

Any model or API

Open or closed, local or hosted. Bring your own keys where you can.

🔓

Open source required

Projects must be open source and fully reproducible. No closed black boxes.

🇦🇲

Armenian encouraged

Not required, but Armenian-language support is a real plus.

📦

Use real data

Anything publicly available, or data we provide before or during the event.

👥

Teams of 3–4

Form your team at kickoff. 15–20 teams in total.

⚖️

Build it here

Come with ideas, not with a finished codebase. The work happens during the sprint.

The 24 hours

Sat 10:00	**Kickoff & briefing** Welcome, theme walkthrough, logistics.
Sat 11:00	**Teams form & plan** Form a team of 3–4, scope your problem and data.
Sat 15:00–18:00	**Mentors on-site** Mentors available in the room — flexible, drop in any time.
Sat → Sun	**Build** Prototype end-to-end early, then harden and evaluate.
Sun 12:00	**Code freeze** Lock the repo, prepare your talk and demo.
Sun 13:00	**Demo Day** 5-minute pitches to the jury.
After demos	**Awards** The jury names the winners.

Your mentors

On-site **Saturday 15:00–18:00** — flexible, drop in any time.



Philipp Guevorguian
Technical Staff · Perceptron AI



Aram Shakhbandaryan
Founder · Empy.ai



Ani Vanyan
ML Researcher · YerevaNN



Khoren Petrosyan
ML Researcher · YerevaNN



Naré Gevorgyan
CEO · AICA



Marie Mikayelyan
CPO · OmniShift



Hovhannes Kuloghlyan
Co-Founder · Wirestock



Armen Grigoryan
Head of TV & Digital Products · Ucom



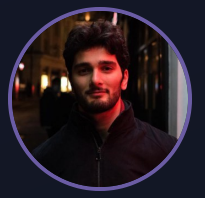
Khachik Badeyan
Technical Portfolio Lead · PMI Science Armenia



Gevorg Minasyan
Async



Davit Karamyan
NVIDIA



Tigran Atoyan
Founder & AI Researcher

How to use your mentors

Come with a specific question

“Is this evaluation sound?” beats “what do you think of our idea?”. Mentors are most useful once you have something concrete.

Show them something running

Even a broken prototype gives a mentor more to work with than a slide.

Ask about scope

The most common 24-hour failure is building too much. Mentors will help you cut.

Product and technical both

Some mentors will push on your model and evaluation, others on who actually uses this and why.

Mentors are on-site Saturday 15:00–18:00. Timing is flexible — grab whoever is free.

What you hand in

🎬

Working demo

It has to run. A rough end-to-end demo beats a polished deck.

📦

Artifact

A repo with reproducible code — plus an app link if you have one.

📊

Evaluation

A test set, rubric, examples, or benchmark — with numbers.

🎯

Impact note

Who uses it, and what changes if it works.

Code freeze is Sunday 12:00 — lock the repo and prepare your pitch before then.

The pitch: 5 minutes, maximum

Five minutes total per team, split however you like between the talk and the demo.

🎤 The talk

The problem, why an LLM is the right tool, how you evaluated it, and what you found — including where it breaks.

🎬 The demo

Show it running on a real case. Live is best; a recording is fine if the network is not.

Split it as you see fit — but at 5:00 you stop. Practise it once before you present.

How you are judged

Five criteria, one rubric for every team — whatever domain you pick.

1. **Idea quality and public-interest fit**
   Is this a real Armenian problem, with a real user and a benefit you can point at?

2. **Working prototype and appropriate LLM use**
   It has to run — and the model has to be doing real work, not decoration.

3. **Evaluation and supporting evidence**
   Numbers, not vibes. Show how you measured it and what came out.

4. **Model limitations, failure modes, and safety**
   Know where it breaks, who it could harm, and what you do about it.

5. **Real-world feasibility and reproducibility**
   Could this survive outside the demo — and can someone else run it?

0 — missing · **3 — adequate** · **5 — exceptional** · **25 points total**

The rubric at a glance

Criterion-specific evidence anchors.

Criterion	0 points	3 points — adequate	5 points — exceptional
1. Idea quality and public-interest fit	Problem is unclear, outside scope, or has no identified user.	A clear Armenia-relevant problem and user are identified; the benefit is plausible.	Local need and user workflow are well understood, with specific and measurable potential impact.
2. Working prototype and appropriate LLM use	No working prototype is shown, or the LLM/VLM is merely decorative.	The core workflow works on a representative case, and the model's role is credible.	The system works end to end across varied cases and shows why an LLM/VLM is better than a simpler alternative.
3. Evaluation and supporting evidence	No meaningful evaluation evidence is provided.	An appropriate test set, examples, metric, or rubric is used and results are reported.	Evaluation is realistic and reproducible, includes a baseline or error analysis, and supports the team's claims.
4. Model limitations, failure modes, and safety	Relevant limitations, failure modes, or safety risks are not identified.	Important risks and failure cases are identified, with basic safeguards or human escalation.	Risky cases are tested systematically and effective mitigation, abstention, escalation, or monitoring is demonstrated.
5. Real-world feasibility and reproducibility	The project relies on toy assumptions, is not reproducible, or has no credible path beyond the demo.	It uses realistic data or labelled stand-ins, documents setup, and describes a plausible deployment path.	It fits real workflows and considers integration, cost, latency, privacy, and maintenance; reproduction is straightforward.

One project, graded three ways

The next five slides grade the same project against each criterion — so you can see exactly what separates a 0, a 3 and a 5.

THE RUNNING EXAMPLE

Guarded symptom triage in Armenian

An assistant that asks about symptoms and routes the person to self-care, a GP, or urgent care. Not diagnosis.

Healthcare is only the vehicle here — the same distinctions apply in every domain.

1 · Idea quality and public-interest fit

Judges are checking that you picked a problem someone in Armenia actually has, that you can name the person who would use it, and that you can say what changes if it works. Vague “AI for good” framing scores nothing here.

0 POINTS

Problem is unclear, outside scope, or has no identified user.

TRIAGE ASSISTANT

“An AI health chatbot for Armenia.” No specific user, no workflow, no stated benefit.

3 — ADEQUATE

A clear Armenia-relevant problem and user are identified; the benefit is plausible.

TRIAGE ASSISTANT

“People outside Yerevan often can’t tell whether a symptom needs a GP or the emergency room.” The user is the patient at home; the benefit is fewer delayed presentations.

5 — EXCEPTIONAL

Local need and user workflow are well understood, with specific and measurable potential impact.

TRIAGE ASSISTANT

The same, plus which regions lack after-hours care, what people do today (call a relative, wait it out), and a target: cut avoidable ER trips at one pilot clinic by a stated share.

2 · Working prototype and appropriate LLM use

Judges will ask you to run it. They are checking that the core loop works on a real case, and that the model earns its place — that a lookup table or a plain form could not have done the same job.

0 POINTS

No working prototype is shown, or the LLM/VLM is merely decorative.

TRIAGE ASSISTANT

Slides and a Figma mock-up. Or: a keyword lookup table, with the model only rewriting the final sentence.

3 — ADEQUATE

The core workflow works on a representative case, and the model's role is credible.

TRIAGE ASSISTANT

A live run: you type "chest tightness since this morning", it asks two follow-up questions in Armenian and routes to urgent care.

5 — EXCEPTIONAL

The system works end to end across varied cases and shows why an LLM/VLM is better than a simpler alternative.

TRIAGE ASSISTANT

Ten varied vignettes run live, including deliberately ambiguous ones — next to a keyword baseline visibly mis-routing the cases the model gets right.

3 · Evaluation and supporting evidence

“It seems to work” is not evidence. Judges want to know what you measured it on, what number came out, and whether they could get that number again themselves.

0 POINTS

No meaningful evaluation evidence is provided.

TRIAGE ASSISTANT

“We tried a few examples and it worked well.”

3 — ADEQUATE

An appropriate test set, examples, metric, or rubric is used and results are reported.

TRIAGE ASSISTANT

60 vignettes labelled against a nurse-authored triage rubric, with agreement reported as a single accuracy figure.

5 — EXCEPTIONAL

Evaluation is realistic and reproducible, includes a baseline or error analysis, and supports the team’s claims.

TRIAGE ASSISTANT

The same 60 vignettes plus a keyword baseline to beat, accuracy broken down by severity, the confusion pairs named — and one script that re-runs every number.

4 · Model limitations, failure modes, and safety

Every system fails somewhere. Judges are checking that you know where it fails, who is hurt when it does, and that you built something to catch it — not that you claim it never fails.

0 POINTS

Relevant limitations, failure modes, or safety risks are not identified.

TRIAGE ASSISTANT

No mention that telling someone with cardiac symptoms to rest at home could kill them.

3 — ADEQUATE

Important risks and failure cases are identified, with basic safeguards or human escalation.

TRIAGE ASSISTANT

Under-triage named as the dangerous direction, a permanent “if in doubt, seek care” fallback, and an explicit “this is not diagnosis” boundary.

5 — EXCEPTIONAL

Risky cases are tested systematically and effective mitigation, abstention, escalation, or monitoring is demonstrated.

TRIAGE ASSISTANT

A deliberate red-flag set is run; the under-triage rate is reported separately from overall accuracy; below a confidence threshold the system abstains and escalates instead of guessing.

5 · Real-world feasibility and reproducibility

Judges are asking whether this could exist outside the demo: whether the data is real, whether someone else can run it from a clean checkout, and whether you have thought about what running it actually costs.

0 POINTS

The project relies on toy assumptions, is not reproducible, or has no credible path beyond the demo.

TRIAGE ASSISTANT

Hard-coded English vignettes, an API key pasted into a notebook, and no README.

3 — ADEQUATE

It uses realistic data or labelled stand-ins, documents setup, and describes a plausible deployment path.

TRIAGE ASSISTANT

Real public triage guidelines, Armenian-language input, a README that runs from a clean checkout, and a stated path: a phone line at one clinic.

5 — EXCEPTIONAL

It fits real workflows and considers integration, cost, latency, privacy, and maintenance; reproduction is straightforward.

TRIAGE ASSISTANT

The same, plus cost and latency per consultation, where symptom data is stored and for how long, who updates the guidelines as they change, and one command that reproduces the evaluation.

The jury

Who you present to on Sunday at 13:00.



Gevorg Mantashyan

First Deputy Minister · Ministry of High-Tech Industry



David Zokhrabyan

Growth · OpenAI



Davit Arakelyan

Chief Business Officer · Eleveight AI



Arik Akhverdyan

Co-Founder & CEO · AI9



Vazgen Hakobjanyan

Founder & CEO · Magical Labs



Vazgen Badalyan

Founder · Evidenso



Erik Arakelyan

Senior Researcher · NVIDIA Armenia



Marie Mikayelyan

CPO · OmniShift



Arto Yeritsyan

CEO & Founder · Async

The organizers

Hack Armenia is run by the Armenia LLM Summer School, AI9 and YerevaNN.



Armen Aghajanyan
Perceptron AI



Aram Galstyan
USC / Amazon



Erik Arakelyan
Senior Researcher · NVIDIA Armenia



Hrant Khachatryan
Founding Director · YerevaNN



Perouz Taslakian
ServiceNow AI Research



Tatev Vardanyan
YerevaNN



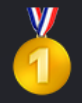
Arik Akhverdyan
Co-Founder & CEO · AI9



Roza Khachatryan
COO · AI9 Startup Campus

Prizes

OpenAI is sponsoring the hackathon prizes with API credits.



1st place

$12,000

in OpenAI API credits



2nd place

$7,500

in OpenAI API credits



3rd place

$5,000

in OpenAI API credits

1st prize


1st PRIZE
$12 000
in OpenAI API credits
OpenAI
AI9 startup campus
YN²

2nd prize


2nd PRIZE
$7500
in OpenAI API credits
OpenAI
AI9 startup campus
YN²

3rd prize


3rd PRIZE
$5 000
in OpenAI API credits
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Supported by

🏛️

Ministry of High-Tech Industry

Hack Armenia is generously supported by the Ministry of High-Tech Industry of the Republic of Armenia.

🤝

ISAA

Supporting the sprint alongside the Ministry.

Special prize

A separate award, presented by Eleveight AI.



$5,000 equivalent GPUs

Eleveight AI is presenting its own special prize at Hack Armenia — GPUs worth $5,000, so the winning team leaves with real compute to keep building.

NOW GO BUILD

Good luck

Ship something that runs, measure it honestly, and tell us who it helps.

Demo Day — Sunday 13:00 · armeniallm@gmail.com

